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Experiment-02

Aim : To containerize application using docker.

This Experiment maps to LO4.

Installation of Docker :

Step 1: Check System Requirements

- Windows 10 (64-bit) Pro, Enterprise, or Education (version 22H2 or later), or Windows 11.
- Hardware virtualization enabled in BIOS/UEFI.
- WSL 2 (recommended) or Hyper-V.

Step 2: Enable Virtualization

1. Restart your computer.
2. Enter the BIOS/UEFI settings (commonly by pressing **F2**, **Delete**, or **Esc** during startup).
3. Enable **Intel VT-x** or **AMD-V**.
4. Save changes and restart Windows.

Step 3: Enable WSL 2

Open **PowerShell as Administrator** and run:

```
wsl --install
```

Restart your computer if prompted.

To verify the installation:

```
wsl --status
```

Step 4: Download Docker Desktop

Download the installer from:

Docker Desktop for Windows

Step 5: Install Docker Desktop

1. Run the installer.
2. Leave **Use WSL 2 instead of Hyper-V** selected (recommended).
3. Click **OK** or **Install**.

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4. Wait for the installation to finish.
5. Restart your computer if prompted.

Step 6: Start Docker Desktop

- Open **Docker Desktop** from the Start menu.
- Accept the license agreement if prompted.
- Wait until the status shows **Docker Desktop is running**.

Step 7: Verify the Installation

Open **Command Prompt** or **PowerShell** and run:

```
docker --version
docker version
docker info
```

Step 8: Test Docker

Run:

```
docker run hello-world
```

If Docker is working correctly, it will download the `hello-world` image and display a confirmation message.

Step 9: Pull an Ubuntu Image

```
docker pull ubuntu
```

List downloaded images:

```
docker images
```

Run an interactive Ubuntu container:

```
docker run -it ubuntu bash
```

What is Docker?

Docker is an open-source containerization platform that enables developers to build, package, distribute, and run applications in isolated environments called **containers**. It includes the application along with all its dependencies, libraries, and configuration files, ensuring that it runs consistently across different systems.

Key Features of Docker

- Lightweight and fast.
- Platform-independent.
- Easy application deployment.
- Efficient resource utilization.
- Supports microservices architecture.

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- Simplifies application development, testing, and deployment.

What is a Docker Image?

A **Docker image** is a read-only template that contains everything needed to run an application, including the application code, runtime, system libraries, dependencies, and configuration files. Images are used to create Docker containers.

Characteristics of Docker Images

- Read-only and immutable.
- Can be shared through Docker Hub or private registries.
- Multiple containers can be created from a single image.
- Built using a Dockerfile.

Examples:

- Ubuntu image
- Nginx image
- MySQL image
- Python image

What is a Docker Container?

A **Docker container** is a running instance of a Docker image. It is an isolated environment where an application executes along with all its required dependencies. Containers share the host operating system's kernel, making them lightweight and faster than traditional virtual machines.

Characteristics of Docker Containers

- Lightweight and portable.
- Starts and stops quickly.
- Isolated from other containers.
- Can be created, started, stopped, restarted, and deleted as needed.
- Multiple containers can run simultaneously on the same host.

Difference Between Docker Image and Docker Container

Docker Image	Docker Container
A read-only template.	A running instance of an image.
Used to create containers.	Executes the application.
Cannot be modified while running.	Can be started, stopped, and removed.
Stored in Docker Hub or local storage.	Runs on the Docker Engine.

One image can create multiple containers. Each container is created from an image.

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Example

- **Docker Image:** Ubuntu
- **Docker Container:** A running Ubuntu environment created using the Ubuntu image.

Docker Command	Description
`docker --version`	Displays the installed Docker version.
`docker version`	Shows Docker client and server version information.
`docker info`	Displays detailed information about Docker system configuration.
`docker help`	Shows a list of available Docker commands and help options.
`docker search <image_name>`	Searches for Docker images on Docker Hub.
`docker pull <image_name>`	Downloads an image from Docker Hub to the local system.
`docker images`	Lists all Docker images stored on the local machine.
`docker image ls`	Alternative command to list Docker images.
`docker run <image_name>`	Creates and starts a container from a Docker image.
`docker run -it <image_name> bash`	Runs a container in interactive mode with a terminal.
`docker run -d <image_name>`	Runs a container in detached mode (background).
`docker run -p <host_port>:<container_port> <image_name>`	Maps a container port to a host machine port.
`docker ps`	Displays currently running containers.
`docker ps -a`	Displays all containers, including stopped containers.
`docker start <container_id>`	Starts a stopped container.
`docker stop <container_id>`	Stops a running container.
`docker restart <container_id>`	Restarts a container.
`docker rm <container_id>`	Removes a container.
`docker rm -f <container_id>`	Forcefully removes a running container.
`docker exec -it <container_id> bash`	Opens a command shell inside a running container.

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`docker logs <container_id>`	Displays logs generated by a container.
`docker inspect <container_id>`	Shows detailed information about a container or image.
`docker stats`	Displays real-time CPU, memory,

Docker Command	Description
	network, and resource usage of containers.
`docker stats --no-stream`	Displays container statistics once without continuous updates.
`docker rmi <image_name>`	Removes a Docker image.
`docker image prune`	Removes unused Docker images.
`docker system df`	Shows Docker disk space usage.
`docker system prune`	Removes unused containers, images, networks, and cache.
`docker build -t <image_name> .`	Builds a Docker image using a Dockerfile.
`docker tag <image> <new_name>`	Creates a new tag for an image.
`docker login`	Logs in to Docker Hub.
`docker logout`	Logs out from Docker Hub.
`docker push <image_name>`	Uploads an image to Docker Hub.
`docker cp <source> <container>:<path>`	Copies files from the host to a container.
`docker cp <container>:<path> <destination>`	Copies files from a container to the host.
`docker network ls`	Lists Docker networks.
`docker network create <name>`	Creates a new Docker network.
`docker volume ls`	Lists Docker volumes.
`docker volume create <name>`	Creates a new Docker volume.
`docker run hello-world`	Tests whether Docker installation is working correctly.

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Output :

```
C:\Users\Lab 203>docker --version
Docker version 29.3.1, build c2be9cc

C:\Users\Lab 203>docker images
                                                                                                    i Info
IMAGE                ID             DISK USAGE   CONTENT SIZE   EXTRA
new_img_name:latest  d0f28c79138d       202MB         53.2MB
nginx:latest         7150b3a39203       240MB         65.8MB    U
redis:latest         009cc37796fb       204MB         55.3MB    U
sample69:latest      9fc8bb8bc4fe      1.57GB          395MB    U
ubuntu:20.04         8feb4d8ca535       111MB         29.3MB
ubuntu:latest        3131b4cc82a7       161MB         45.3MB    U

C:\Users\Lab 203>docker search ubuntu
NAME                            DESCRIPTION                                     STARS     OFFICIAL
ubuntu                          Ubuntu is a Debian-based Linux operating sys…   17859     [OK]
ubuntu/squid                    Squid is a caching proxy for the Web. Long-t…   129
ubuntu/nginx                    Nginx, a high-performance reverse proxy & we…   141
ubuntu/bind9                    BIND 9 is a very flexible, full-featured DNS…   120
ubuntu/kafka                    Apache Kafka, a distributed event streaming …   62
ubuntu/zookeeper                ZooKeeper maintains configuration informatio…   15
ubuntu/apache2                  Apache, a secure & extensible open-source HT…   104
ubuntu/prometheus               Prometheus is a systems and service monitori…   80
ubuntu/jre                      Chiseled Java runtime based on Ubuntu. Long-…   25
ubuntu/dotnet-aspnet            Chiselled Ubuntu runtime image for ASP.NET a…   26
ubuntu/python                   A chiselled Ubuntu rock with the Python runt…   33
ubuntu/mysql                    MySQL open source fast, stable, multi-thread…   74
ubuntu/postgres                 PostgreSQL is an open source object-relation…   45
ubuntu/dotnet-runtime           Chiselled Ubuntu runtime image for .NET apps…   24
ubuntu/redis                    Redis, an open source key-value store. Long-…   26
```

```
C:\Users\Lab 203>docker search ubuntu --filter stars=1000
NAME      DESCRIPTION                                     STARS     OFFICIAL
ubuntu    Ubuntu is a Debian-based Linux operating sys…   17859     [OK]

C:\Users\Lab 203>docker search ubuntu --filter stars=100
NAME             DESCRIPTION                                     STARS     OFFICIAL
ubuntu           Ubuntu is a Debian-based Linux operating sys…   17859     [OK]
ubuntu/squid     Squid is a caching proxy for the Web. Long-t…   129
ubuntu/nginx     Nginx, a high-performance reverse proxy & we…   141
ubuntu/bind9     BIND 9 is a very flexible, full-featured DNS…   120
CONTAINER ID   NAME         CPU %     MEM USAGE / LIMIT   MEM %     NET I/O   BLOCK I/O   PIDS
1647b54df784   container3   0.00%     0B / 0B             0.00%     0B / 0B   0B / 0B     0


C:\Users\Lab 203>docker rename container2 container3
failed to rename container: Error response from daemon: No such container: container2

C:\Users\Lab 203>docker rename container3 container2

C:\Users\Lab 203>docker pull nginx
Using default tag: latest
latest: Pulling from library/nginx
Digest: sha256:5a88c9c45479443d7be2eadc894b4ed0a9801bae03d97a5760ae13b5c2005942
Status: Image is up to date for nginx:latest
docker.io/library/nginx:latest
```

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```
C:\Users\Lab 203>docker info
Client:
 Version:    29.3.1
 Context:    desktop-linux
 Debug Mode: false
 Plugins:
  agent: Docker AI Agent Runner (Docker Inc.)
    Version:  v1.32.4
    Path:     C:\Program Files\Docker\cli-plugins\docker-agent.exe
  ai: Docker AI Agent - Ask Gordon (Docker Inc.)
    Version:  v1.20.1
    Path:     C:\Program Files\Docker\cli-plugins\docker-ai.exe
  buildx: Docker Buildx (Docker Inc.)
    Version:  v0.32.1-desktop.1
    Path:     C:\Program Files\Docker\cli-plugins\docker-buildx.exe
  compose: Docker Compose (Docker Inc.)
    Version:  v5.1.0
    Path:     C:\Program Files\Docker\cli-plugins\docker-compose.exe
  debug: Get a shell into any image or container (Docker Inc.)
    Version:  0.0.47
    Path:     C:\Program Files\Docker\cli-plugins\docker-debug.exe
  desktop: Docker Desktop commands (Docker Inc.)
    Version:  v0.3.0
    Path:     C:\Program Files\Docker\cli-plugins\docker-desktop.exe
  dhi: CLI for managing Docker Hardened Images (Docker Inc.)
    Version:  v0.0.0-alpha
    Path:     C:\Program Files\Docker\cli-plugins\docker-dhi.exe
  extension: Manages Docker extensions (Docker Inc.)
    Version:  v0.2.31
    Path:     C:\Program Files\Docker\cli-plugins\docker-extension.exe
  init: Creates Docker-related starter files for your project (Docker Inc.)
    Version:  v1.4.0
    Path:     C:\Program Files\Docker\cli-plugins\docker-init.exe
  mcp: Docker MCP Plugin (Docker Inc.)
    Version:  v0.40.1
    Path:     C:\Program Files\Docker\cli-plugins\docker-mcp.exe
  model: Docker Model Runner (Docker Inc.)
    Version:  v1.1.5
    Path:     C:\Program Files\Docker\cli-plugins\docker-model.exe
  offload: Docker Offload (Docker Inc.)
```

```
C:\Users\Lab 203>docker --help
Usage:  docker [OPTIONS] COMMAND

A self-sufficient runtime for containers

Common Commands:
  run         Create and run a new container from an image
  exec        Execute a command in a running container
  ps          List containers
  build       Build an image from a Dockerfile
  bake        Build from a file
  pull        Download an image from a registry
  push        Upload an image to a registry
  images      List images
  login       Authenticate to a registry
  logout      Log out from a registry
  search      Search Docker Hub for images
  version     Show the Docker version information
  info        Display system-wide information

Management Commands:
  agent*      Docker AI Agent Runner
  ai*         Docker AI Agent - Ask Gordon
  builder     Manage builds
  buildx*     Docker Buildx
  compose*    Docker Compose
  container   Manage containers
  context     Manage contexts
  debug*      Get a shell into any image or container
  desktop*    Docker Desktop commands
  dhi*        CLI for managing Docker Hardened Images
  extension*  Manages Docker extensions
```

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```
C:\Users\Lab 203>docker start 1647b54df784
1647b54df784

C:\Users\Lab 203>docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES

C:\Users\Lab 203>docker ps -a
CONTAINER ID   IMAGE          COMMAND                  CREATED          STATUS                       PORTS                    NAMES
0b6fa580b2b6   ubuntu:20.04   "/bin/bash"              4 minutes ago    Exited (0) 4 minutes ago                              silly_brown
1647b54df784   ubuntu:latest  "/bin/bash"              21 minutes ago   Exited (0) About a minute ago                         container2
66097bfabcdf   ubuntu         "/bin/bash"              20 hours ago     Exited (255) 43 minutes ago                           tender_payne
ed5b6c3b864e   ubuntu         "/bin/bash"              20 hours ago     Exited (0) 20 hours ago                               hopeful_grothendieck
6684d45157fa   sample69       "docker-entrypoint.s…"   3 months ago     Exited (255) 20 hours ago    0.0.0.0:3000->3000/tcp   kind_galileo
79d421376f82   redis          "docker-entrypoint.s…"   3 months ago     Exited (255) 20 hours ago    6379/tcp                 bold_jennings
c49b0c44e8c3   186072bba1b2   "bash"                   3 months ago     Exited (2) 3 months ago                               heuristic_beaver
7ac1bd2e0714   7150b3a39203   "/docker-entrypoint.…"   3 months ago     Created                                               stupefied_poitras
68bb2a68ae5a   7150b3a39203   "/docker-entrypoint.…"   3 months ago     Exited (255) 20 hours ago    0.0.0.0:8080->80/tcp     beautiful_hugle
e8d2152bd95e   7150b3a39203   "/docker-entrypoint.…"   3 months ago     Exited (255) 20 hours ago    80/tcp                   elated_wescoff
8fa73e36d210   7150b3a39203   "/docker-entrypoint.…"   3 months ago     Exited (0) 3 months ago                               upbeat_mahavira
```

```
C:\Users\Lab 203>docker run --name container10 sample69:latest
Server running on port 3000

C:\Users\Lab 203>docker rm container10
container10

C:\Users\Lab 203>docker run hello-world
Unable to find image 'hello-world:latest' locally
latest: Pulling from library/hello-world
4f55086f7dd0: Pull complete
d5e71e642bf5: Download complete
Digest: sha256:c3cbe1cc1aa588a64951ac6286e0df7b27fe2e6324b1001c619bb358770c0178
Status: Downloaded newer image for hello-world:latest

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
 1. The Docker client contacted the Docker daemon.
 2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
    (amd64)
 3. The Docker daemon created a new container from that image which runs the
    executable that produces the output you are currently reading.
 4. The Docker daemon streamed that output to the Docker client, which sent it
    to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
 $ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
 https://hub.docker.com/

For more examples and ideas, visit:
 https://docs.docker.com/get-started/
```

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```
C:\Users\Lab 203>docker network ls
NETWORK ID     NAME      DRIVER    SCOPE
fab7eb6309a9   bridge    bridge    local
0ee86bc24b4d   host      host      local
6d6bc86843ec   none      null      local

C:\Users\Lab 203>docker volume ls
DRIVER    VOLUME NAME

C:\Users\Lab 203>docker exec -it container2 sh
Error response from daemon: container 1647b54df784cb3ea10778c1c9a4d9dd33ef9f7090b23c6f6aa9ca65773202c3 is not running

C:\Users\Lab 203>docker start 0b6fa580b2b6
0b6fa580b2b6

C:\Users\Lab 203>docker exec -it 0b6fa580b2b6 sh
Error response from daemon: container 0b6fa580b2b69abcc1bd578edd306c54d4f52ed16deada0be55a988e17d065e4 is not running

C:\Users\Lab 203>docker stop 0b6fa580b2b6
0b6fa580b2b6

C:\Users\Lab 203>docker start 1647b54df784
1647b54df784

C:\Users\Lab 203>docker exec -it 0b6fa580b2b6 sh
Error response from daemon: container 0b6fa580b2b69abcc1bd578edd306c54d4f52ed16deada0be55a988e17d065e4 is not running

C:\Users\Lab 203>docker image ls

IMAGE                 ID             DISK USAGE   CONTENT SIZE   EXTRA
hello-world:latest    c3cbe1cc1aa5       25.9kB         9.49kB   U
new_img_name:latest   d0f28c79138d        202MB         53.2MB
nginx:latest          5a88c9c45479        241MB           66MB
redis:latest          009cc37796fb        204MB         55.3MB   U
sample69:latest       9fc8bb8bc4fe       1.57GB          395MB   U
ubuntu:20.04          8feb4d8ca535        111MB         29.3MB   U
ubuntu:latest         3131b4cc82a7        161MB         45.3MB   U
```

Conclusion : From this experiment, I understood that Docker is a containerization platform used to create, manage, and run applications in isolated environments. I learned how to install Docker, verify its installation, and use basic Docker commands.

I understood that Docker images are templates that contain the application and its required dependencies, while containers are running instances created from those images. I learned how to search, download, and run Docker images using commands like `docker search`, `docker pull`, and `docker run`.

I also learned how to manage containers using commands such as `docker ps`, `docker stop`, and `docker stats` to monitor container performance. The port mapping command `docker run -p` helps to access applications running inside containers from the host machine.